CHAPTER 16

Excitons Under the Influence of External Fields

In this chapter we discuss the behaviors of excitons under the influence of external fields such as magnetic and electric field.

16.2 The Effects of Electric Field

If apply a static electric field $\boldsymbol{E}_s$ on the exciton resonances in semiconductors:

- There will be a Stark effect.
- The influence of Es on the band-to-band transitions, on the continuum states is also known as the Franz–Keldysh effect- modification on the absorption spectrum - resulting in a red-shift of the band-to-band absorption tail
- Influence the refractive index in the transparent spectral region below the exciton resonances, resulting in Pockels- or Kerr-effect-like phenomena.
- The influence of $\boldsymbol{E}_s$ on the eigenfrequencies.

16.2.1 Bulk Semiconductors

It is not easy to observe the Stark effect for excitons in three-dimensional. To get an observable shift of the eigenstates, the electric field is in the order of 10^6 Vm^{-1}

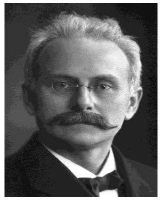
Such fields broaden or even destroy the exciton resonances due to two effects.

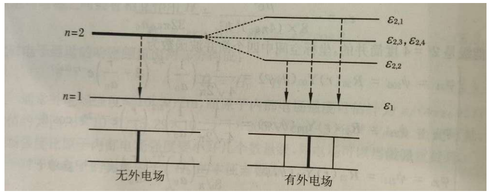
- One is field ionization of the exciton by tunneling through the finite Coulomb barrier at finite fields (Fig.16.10)
- The other is impact ionization.

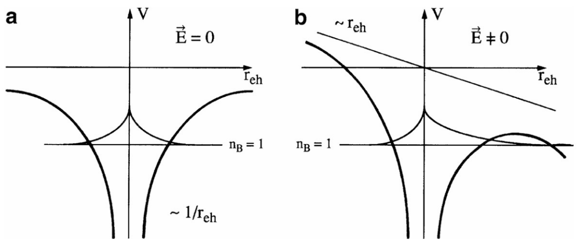
Stark Effect

The Stark Effect refers to the splitting of spectral lines of atoms when exposed to an external static electric field, arising from the perturbation of atomic energy levels by the electric field

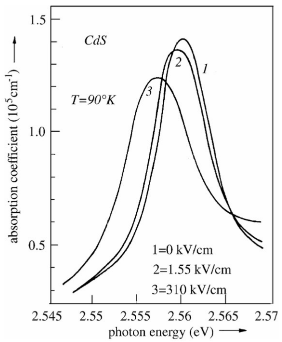
- First observed in 1913 by the German physicist Johannes Stark, who detected spectral line splitting while studying the hydrogen atom spectrum
- Similar to the Zeeman Effect (spectral splitting caused by magnetic fields), the Stark Effect specifically resulting from electric field-induced changes to atomic energy levels
- Atomic electron energy levels initially exhibit "degeneracy"—different quantum states correspond to the same energy value
- An external static electric field breaks this degeneracy, splitting originally overlapping energy levels into multiple sub-levels with distinct energies
- Linear Stark Effect: The degree of splitting is proportional to the strength of the external electric field. It is commonly observed in low energy levels of simple systems like hydrogen atoms
- Quadratic Stark Effect: The splitting degree is proportional to the square of the electric field strength. It typically occurs in strong electric field environments or complex atomic systems







The Coulomb potential of a three-dimensional exciton without (a) and with (b) an applied constant electric field



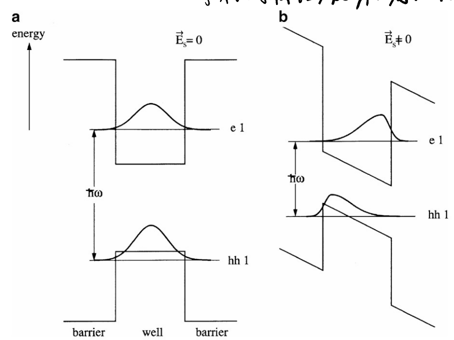
Influence of an external field on the absorption spectrum of the B, n_B = 1 exciton of CdS for various applied electric fields

16.2.2 Stark Effect in Semiconductor Structures of Reduced Dimensionality-QCSE

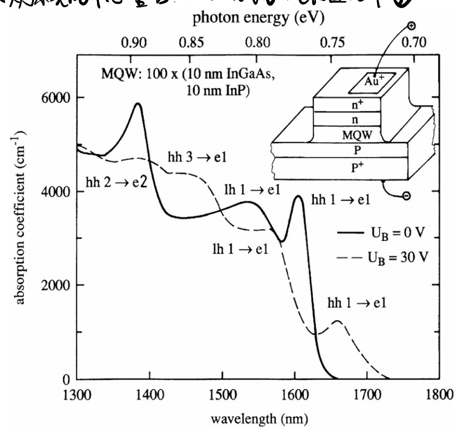
By confining the electron and hole between barriers, e.g., in quantum wells, wires or dots can overcome the problem of field ionisation in bulk semiconductors. Three effects:

- The red-shift of the exciton. 第一子带电子能级↓ ~ 空穴 ~↑，能量差减小
- The decrease of its oscillator strength. e.h波函数从原先的中心重合 变为分离，跃迁几率↓
- The appearance of forbidden transitions.

宇称对称性破坏，有向失效

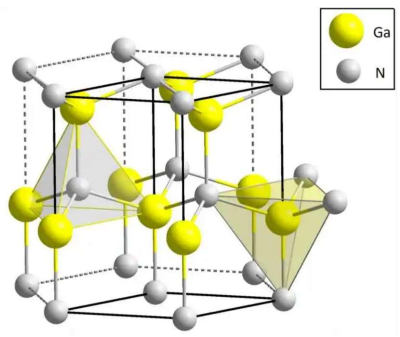


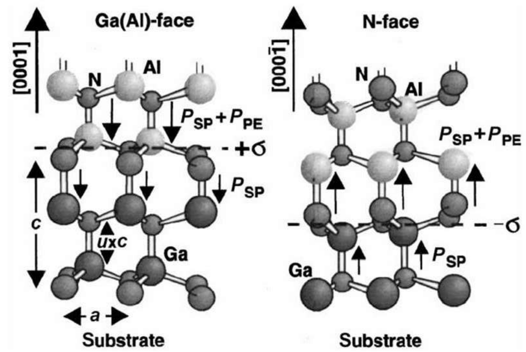
The bandstructure of a quantum well without (a) and with (b) an applied constant electric field perpendicular to the wells, but neglecting the Coulomb interaction between electrons and holes



The quantum-confined Stark effect in InGaAs/InP MQW samples

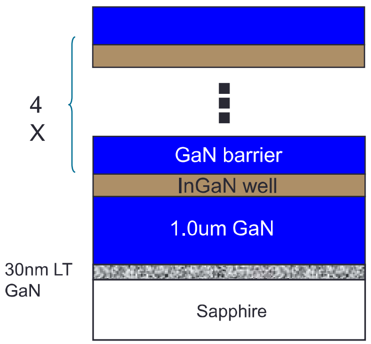
Piezoelectric (PE) and spontaneous polarization (SP) effects in GaN





Piezoelectric (PE) and spontaneous polarization (SP) effects in Ga(Al) or N face AlGaN/GaN heterostructures

QCSE in InGaN/GaN QWs

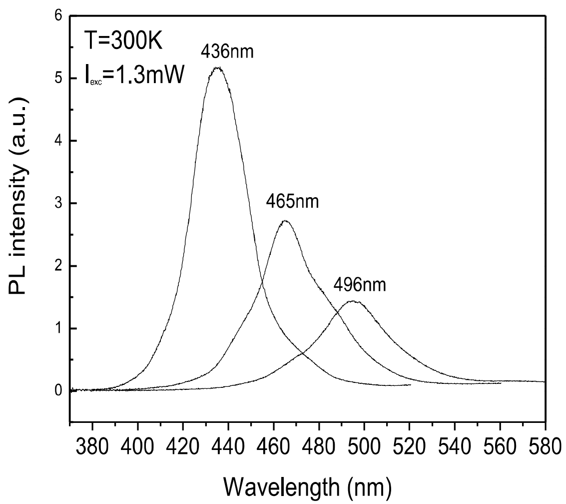


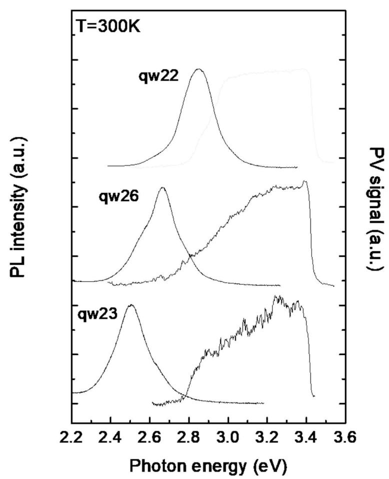
Schematic diagram of sample structure

Sample Parameters

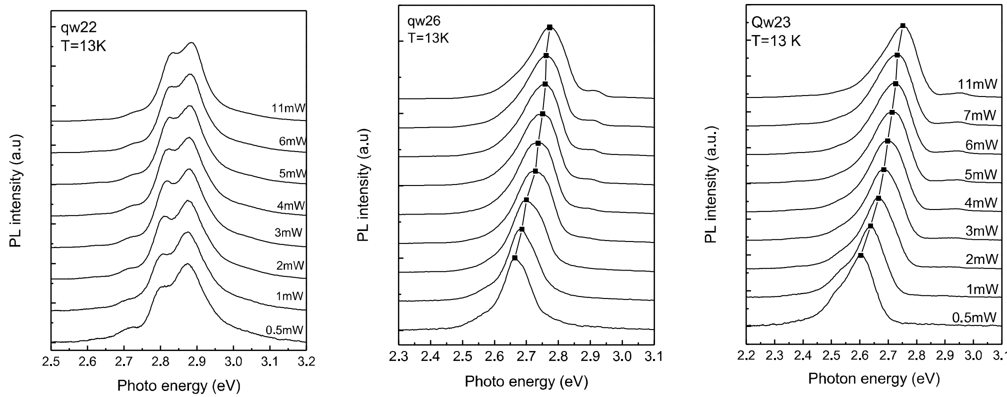
Sample	Well thickness (Å)	Barrier thickness (Å)	Indium fraction
qw22	22	176	0.19
qw26	33	176	0.19
qw23	44	176	0.19
qw29	44	176	0.13
qw27	44	176	0.09

PL and PV Spectra measurement

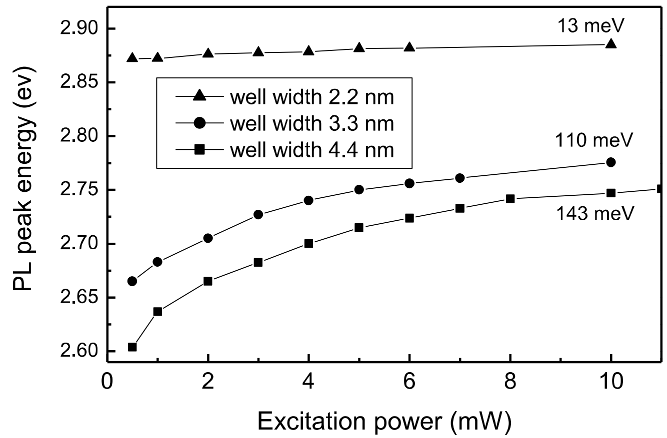




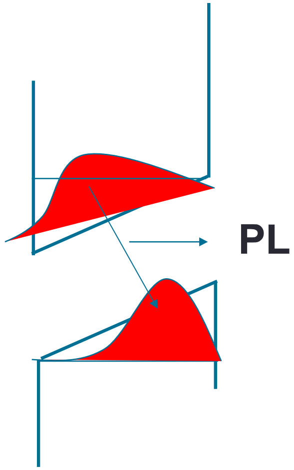
PL Spectra-Excitation Intensity Dependence



Excitation Intensity Dependence

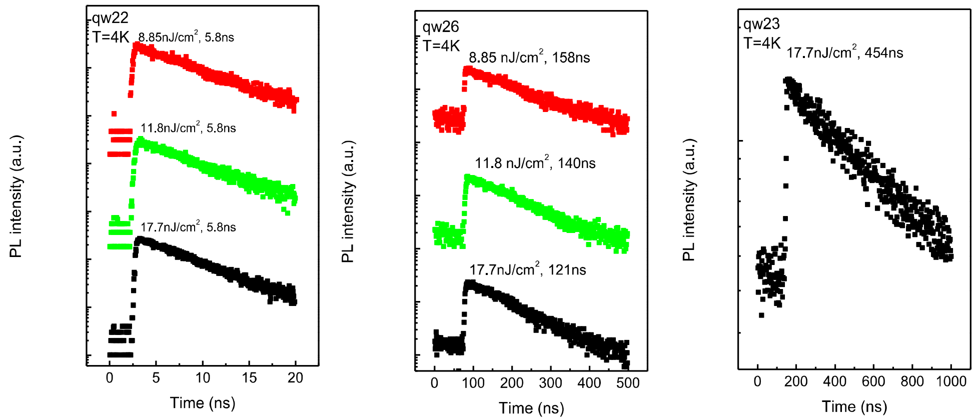


Quantum Confined Stark Effect



Schematic view of optical transitions in the tilt InGaN quantum wells

PL Decay - Well Width Dependence



- The built-in electric fields separate the electrons and holes spatially, reduce the oscillator strength and thus recombination rate, therefore long decay time is anticipated
  - The PL decay time will increase with the well width
  - The PL decay time will be excitation intensity dependent

16.3 Strain Fields on the

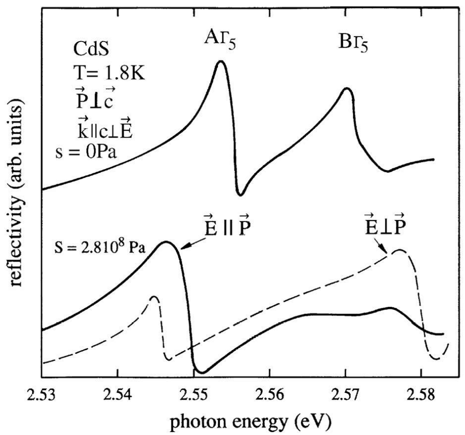
- The dependence of the energy of the band extrema on the strain, the so-called deformation potential is defined for conduction and valance band by

$$\Xi = a\frac{\mathrm{d}E_{c,v}}{\mathrm{d}x}$$

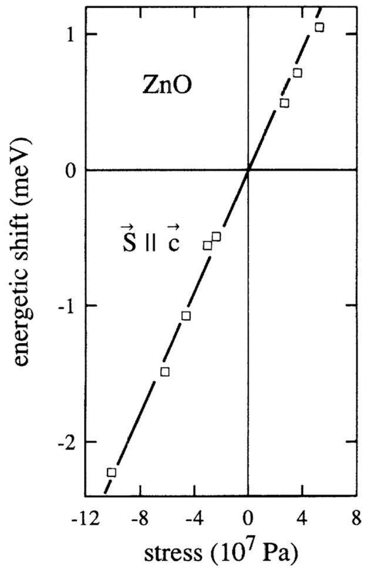
where *a* is the lattice constant, $dE_{c,v}$ is the shift of the band edge caused by a length variation of the unit cell by an amount $\mathrm{d}x$. Deformation potentials are defined for various types of strain like hydrostatic, uniaxial or shear strain

- For excitons the relevant quantity is the sum effects of the deformation potentials on conduction and valence bands. As in the case of electric field, an applied stress changes the eigenenergies of the exciton states and may also lead to a splitting of degenerate states,

16.3.1 Bulk Semiconductors



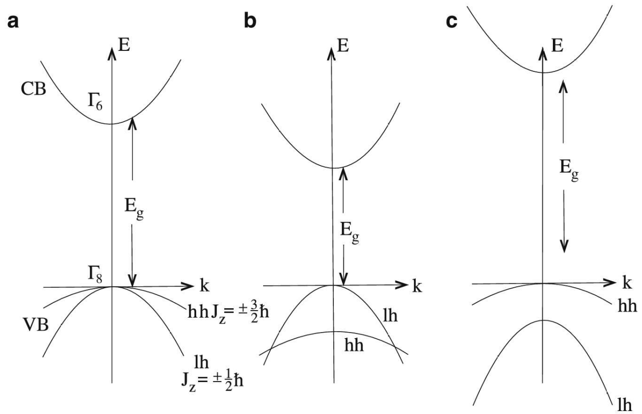
The excitonic reflection spectra of CdS under uniaxial stress



The shift of a BEC luminescence line in ZnO under the influence of strain

16.3 Strain Fields on the Structures of Reduced Dimensionality

Strain-induced splitting of the Γ_8 valence band is of special importance for strained quantum wells and superlattices whenever a layer is grown with a lattice constant different from that of the substrate.



Schematic of the influence of strain on the alignment of the hh and lh valence bands and on the value of the band gap (a) zero strain (b) two-dimensional tensile strain and normal uniaxial compressive strain (c) and two-dimensional compressive strain and normal uniaxial tensile strain.